Curriculum vitae

Ferran Alía Castillejos

Data science and physical engineering, UPC Barcelona

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Now

Fourth year at UPC. This summer I was at TU Dresden with the Physics of Life group, simulating microscope optics. Interested in research and internship positions in CFD, HPC, physics-informed machine learning or surrogate modeling.

Experience

Since 2026 — **Contributor, Towards Data Science**
Long-form articles since March 2026. Most of them come out of the projects below.

Jun – Aug 2026 — **Research intern, Physics of Life, TU Dresden**
Simulating electromagnetic propagation through specimen and objective, as virtual instrumentation for learned brightfield imaging. On the side, a GNN surrogate for multiple scattering between beads.

Jun – Aug 2025 — **Machine learning intern, MLCode**
Built an automated framework for evaluating and benchmarking retrieval-augmented generation systems.

Since 2022 — **Volunteer mentor**
Computer science and robotics workshops: Linux, 3D printing, programming.

2022 – 2023 — **Robotics teacher, Punt Multimèdia**
Taught Arduino, micro:bit, and 3D printing and modeling.

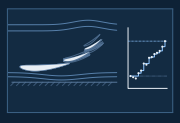
Education

2023 – 2028 — **BSc Data Science & Engineering, UPC**
In the fourth year of five. 8.75/10 average across both degrees.

2023 – 2028 — **BSc Physical Engineering, UPC**
Taken in parallel with the data science degree rather than after it, on the CFIS dual-degree program.

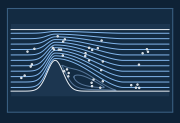
Selected projects

2026



Aerodynamic optimization of a Formula 1 front wing
A surrogate model searched front-wing geometries on a fixed budget of solver runs rather than calling CFD at every step. The wing came out about 15% better on lift-to-drag.
Group project. Python, OpenFOAM, SLURM, MareNostrum V

2026



A PINN for blood flow through a narrowed artery
From 40 velocity readings with 7% noise, a physics-informed network reconstructs the flow, the pressure and the wall shear stress in a 2D artery with a 50% stenosis, and learns the unknown viscosity to within 1–8%. With 20 sensors it settles on the wrong value while the loss curves look fine.
Python, PyTorch

2026



A graph network for multiple scattering in microscopy
A GNN surrogate for the bead-bead coupling in Köhler-illuminated microscope images, fast enough to run in real time. Built at TU Dresden alongside the main internship project.

2026



Giving a language model a persona

A system prompt against three formats of fine-tuning data on Qwen3-4B-Instruct, all with LoRA at r=16, so the differences belong to the data.

Python, PyTorch, Transformers, LoRA

2025

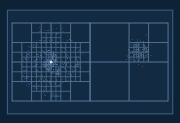


Predicting fetal distress from cardiotocography

Built with Hospital Sant Joan de Déu in a team of nine. Separate encoders over CTG traces, tabular clinical context and ultrasound, at around 0.78 AUC.

Python, PyTorch

2025



Parallel N-body simulation

Direct $O(n^2)$ summation and a Barnes-Hut O(n log n) tree code in C++ with OpenMP. At N = 10,000 direct summation ran about 33× faster on 112 threads than on one; the tree code never got past roughly 7×.

Pair project. C++, OpenMP

2025

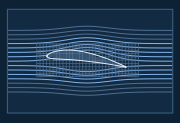


Parallel lattice Boltzmann solver

A D2Q9 solver with BGK collision and a Zou-He inlet in around 200 lines of C++: flow past a cylinder on a 400×400 lattice. Memory-bound, so how you move the data dominates the scaling.

Pair project. C++, OpenMP, MareNostrum 5

2025

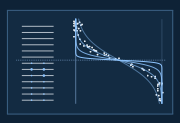


Fixed-wing bird flight simulation

A 3D Navier-Stokes solver written from scratch in NumPy, run on a bird's wing modeled in Blender and checked against OpenFOAM's icoFoam.

Python, NumPy, OpenFOAM, Blender

2024



Fermi-Dirac statistics by Monte Carlo

A Markov chain over the microstates of a fermion system, recovering the Fermi-Dirac occupancy numerically rather than analytically.

Python, NumPy

Figures are schematic illustrations of each problem, not results.

Skills

Languages	Python, C++, R, Haskell, MATLAB
Scientific	OpenFOAM, OpenMP, NumPy, PyTorch, Blender, CAD
Tools	Linux, Git
Methods	CFD, HPC & SLURM, physics-informed ML, Monte Carlo, surrogate modeling
Spoken	Catalan (native), Spanish (native), English (professional), Arabic (beginner)

Awards

Nov 2024, 2025	Participant, Datathon FME
Jun 2025	Physics in Ljubljana, summer school at the University of Ljubljana
Since 2023	Cum laude in several subjects
2023	Admitted to CFIS on a grant, to read the two degrees at once. It takes a small cohort each year by its own entrance exam in mathematics and physics
2019 – 2023	Graduated from high school with honors, and represented it at the Barcelona *Mostra de Recerca Jove*, after UPC's summer courses in algorithms and programming